Result 1 (inverse equivalence evaluation). [mathlib, Equiv.symm_apply_apply]
Let A and B be types. Let e be an equivalence from A to B. Let y be an element of B. Then

$$e^{-1}(y) = e^{-1}(y).$$

Adapted from the cited source.

Result 2 (composed equivalence evaluation). [mathlib, Equiv.trans_apply]
Let A, B and C be types. Let e be an equivalence from A to B. Let f be an equivalence from B to C. Let x be an element of A. Then

$$(f \circ e)(x) = f(e(x)).$$

Adapted from the cited source.

Result 3 (composed equivalence round trip). [mathlib, Equiv.symm_apply_apply]
Let A, B and C be types. Let e be an equivalence from A to B. Let f be an equivalence from B to C. Let x be an element of A. Then

$$(f \circ e)^{-1}((f \circ e)(x)) = x.$$

Adapted from the cited source.

LLM-assisted statements and proofs (gpt-6, 2026-11-07). See PROVENANCE.md. Composition applies the first equivalence and then the second. Both constructions, including their inverse-law evidence, are authored in Clean.